

To Smooth or Not? When Label Smoothing Meets Noisy Labels

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01 Real-world scenario: training with messy labels

- **Messy labels:** Labels are messy in the wild and cross entropy overfits noise while heavy smoothing underfits.
- The goal is to choose smoothing so confidence stays high on correct predictions and low on wrong ones.
- A principled schedule for the rate helps maintain **robust confidence** under imperfect labels.

02 Problem setting and decision rule

- **Decision rule:** Use **LS** when estimated noise $<$ threshold; switch to **NLS** once noise crosses it.
- Binary: $r_{\text{opt}} = r^* - 2\epsilon$. Multi-class: $r_{\text{opt}} = ((K-1)r^* - K\epsilon) / ((K-1) - K\epsilon)$.
- Warm up with **CE**, then anneal r to target; cap $|r|$ for stability.

03 Motivation and thesis: when smoothing should go negative

- High noise implicitly oversmooths predictions and pushes the optimal rate negative.
- A **phase transition** from positive smoothing to negative smoothing appears as noise grows.
- Negative smoothing restores **confidence on true classes** without amplifying noise.

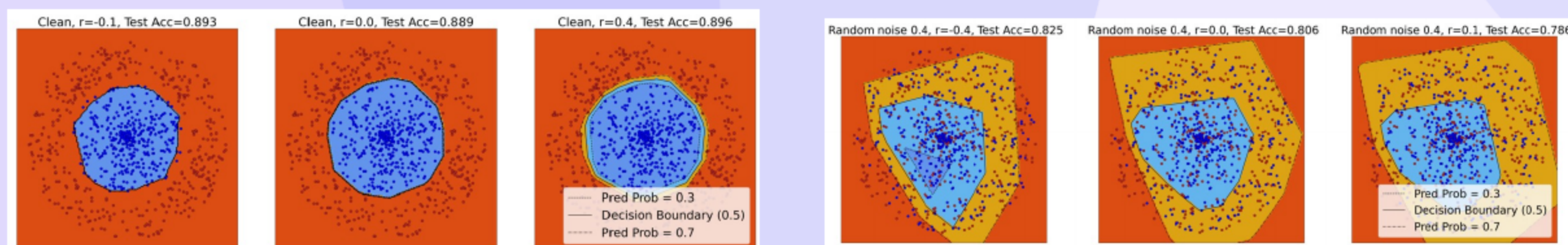
04 Noisy Labels and Generalized Label Smoothing (GLS)

- **Generalized label smoothing** mixes the hard label with a uniform vector using a tunable rate.
- Negative values define negative smoothing and allow extended labels that still sum to one.
- This connects to **loss correction**, **peer loss**, and **complementary labels** under clean assumptions.

05 Confidence Measures We Track

- Cross-entropy **confidence margin** increases with the predicted probability.
- Prediction **confidence gap** = target minus non-target probability.
- Loss-based confidence is **monotonic** in the correct-class probability.

06 How r and Noise Shape Confidence (Intuition First)



- On clean data moderate positive smoothing works best and overly negative values overfit.
- Under symmetric noise decreasing the rate counteracts oversmoothing and improves confidence.
- At higher noise, NLS yields **higher correct** and lower wrong confidence.

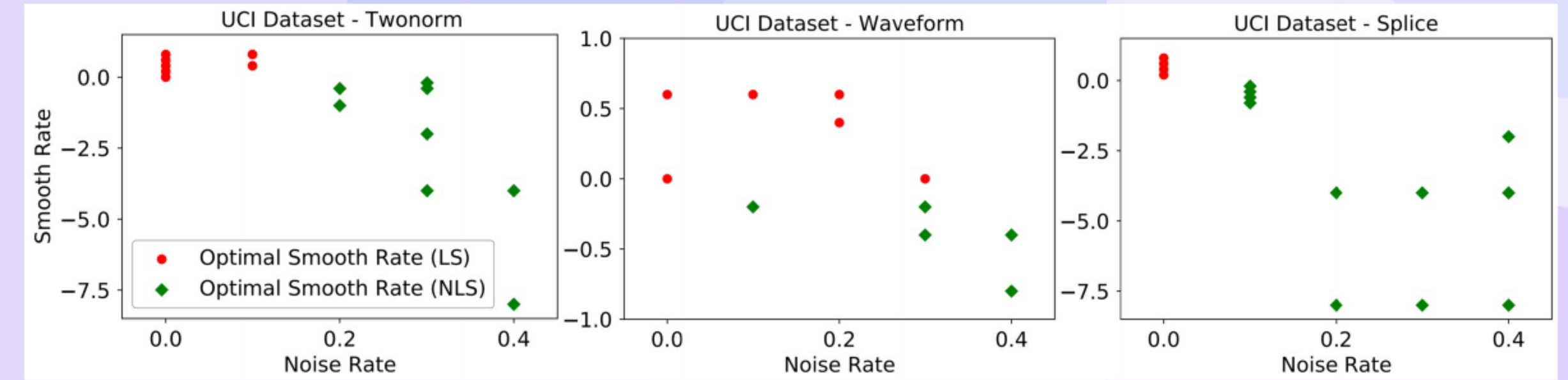
07 Risk Decomposition: From Noisy GLS to Confidence Bias

- Noisy-GLS risk = clean-GLS risk at r^* + two **confidence-bias** terms.
- The bias terms directly raise or lower the confidence margins under label noise.
- This view explains **confidence shifts** when tuning the smoothing rate under noise.

08 Symmetric Noise: Phase Transition in Optimal r

- Binary symmetric: $r_{\text{opt}} = r^* - 2e \Rightarrow$ threshold at $e = r^*/2$.
- Under symmetric noise one bias term vanishes and the remaining term shifts confidence.
- The optimal rate decreases with noise and crosses zero at a noise dependent threshold.
- Below the threshold positive smoothing is preferred and above it **negative smoothing** is preferred.

09 Dedicated: Most surprising finding—optimal r turns negative as noise grows.



- A clear phase transition shows the optimal rate turning negative as label noise increases.
- Negative smoothing undoes noise induced oversmoothing and restores useful confidence.
- This finding provides a **simple rule** to switch from LS to NLS with rising noise.
- Threshold: $r_{\text{opt}} = r^* - 2e$ (binary), crosses zero at $e = r^*/2$

10 Asymmetric noise: class-wise confidence imbalance

- Under **asymmetric noise**, a single global rate can cancel one bias term in the binary case.
- A remaining bias term shifts confidence differently across classes under imbalance.
- Tuning r trades off **class-wise confidence**.

11 Multi-class symmetric noise: threshold formula and intuition

- Under **multi-class symmetry** the optimal rate moves smoothly with the noise level.
- Crossing a noise dependent threshold flips from label smoothing to negative smoothing.
- Decreasing the rate counteracts implicit smoothing and **restores confidence** on true classes.

- **Warm up** with cross entropy and then anneal the rate toward a negative target.
- Use cues such as noise estimates validation stalls and loss diagnostics to trigger switching.
- Linear annealing avoids abrupt shifts and [stabilizes training](#) when noise is high.

Method	CIFAR-10, Symmetric			CIFAR-10, Asymmetric		CIFAR-100, Symmetric	
	$\epsilon = 0.2$	$\epsilon = 0.4$	$\epsilon = 0.6$	$\epsilon = 0.2$	$\epsilon = 0.3$	$\epsilon = 0.4$	$\epsilon = 0.6$
Cross Entropy	86.45	82.72	74.04	88.59	86.14	48.20	38.27
Bootstrap (Reed et al., 2014)	86.06	81.65	75.26	87.69	85.51	47.28	35.81
FLC (Patrini et al., 2017)	84.85	84.98	73.97	89.42	88.25	53.04	41.59
SCE (Wang et al., 2019)	89.39	80.31	75.28	88.07	85.93	49.34	38.87
APL (Ma et al., 2020)	88.42	81.27	76.62	88.75	87.41	51.63	42.31
Peer Loss (Liu & Guo, 2020)	90.21	86.40	79.64	91.38	89.65	62.16	53.72
ELR (Liu et al., 2020)	92.57	91.32	88.86	93.48	92.21	68.03	60.49
AUM (Pleiss et al., 2020)	91.52	87.85	81.71	92.17	90.63	59.29	44.05
Label Smoothing (LS) (Lukasik et al., 2020)	90.24	83.78	75.01	90.61	88.04	55.17	41.63
Negative/Not Label Smoothing (NLS)	89.05	84.85	77.82	90.02	88.42	58.47	46.58

- Label smoothing and negative smoothing are lightweight and effective under rising noise.
- State of the art robust methods are stronger in extreme noise but add complexity.
- Negative smoothing complements advanced methods and narrows gaps in [high noise](#).

- On UCI datasets the predicted optimal rate matches the **clean risk minimizer** under symmetric noise.
- As noise increases negative smoothing becomes increasingly competitive and often best.
- The observed transition aligns with the theory and yields [consistent gains](#).

Smooth Rate	CIFAR-10 Symmetric			
	$\epsilon = 0.0$	$\epsilon = 0.2$	$\epsilon = 0.4$	$\epsilon = 0.6$
$r = 0.8$	92.91±0.06	88.88±1.61	81.48±2.91	73.16±0.16
$r = 0.6$	92.33±0.09	87.50±1.31	82.11±0.86	73.59±0.15
$r = 0.4$	93.05±0.04	87.13±0.07	81.50±1.42	74.21±0.19
$r = 0.0$	91.44±0.16	85.08±0.86	80.42±2.29	75.34±0.13
$r = -0.4$	93.55±0.06	87.55±0.08	81.58±0.19	75.95±0.13
$r = -0.8$	92.74±0.05	88.46±0.11	81.56±0.15	76.15±0.14
$r = -1.0$	92.58±0.08	88.58±0.08	81.95±0.10	76.20±0.10
$r = -2.0$	93.30±0.03	88.78±0.09	83.64±0.15	76.11±0.07
$r = -4.0$	93.13±0.04	88.90±0.07	84.34±0.13	77.22±0.09
$r = -6.0$	93.14±0.08	88.94±0.11	84.52±0.13	77.42±0.16
$p\text{-value} =$	0.0004	0.008	0.011	$< 1e - 14$

- On CIFAR negative smoothing leads as noise increases with strong improvements at high noise.
- Under both symmetric and asymmetric flips negative smoothing remains competitive.
- The accuracy gains are [statistically significant](#) in many high noise settings.

16

Evidence on CIFAR: confidence behavior on test

- Under noise negative smoothing raises confidence for correct predictions and lowers it for wrong ones.
- **Confidence distributions** reflect the intended separation between correct and incorrect cases.
- This separation produces **clearer margins** and aligns with the bias view.

17

Connections: loss correction, peer, and complementary labels as NLS (when/limits).

- **Equivalences:** Loss correction reduces to negative smoothing under symmetric noise with matching rates.
- Peer loss equals negative smoothing when class proportions are balanced in the training data.
- Complementary label training corresponds to an extreme negative smoothing limit.

18

Practical significance on real-world noisy datasets

Results of baselines are obtained through the public leader board.

Method	Clothing 1M	CIFAR-10N Aggre	CIFAR-10N Rand1	CIFAR-10N Worse	CIFAR-100N Fine
CE	68.94	87.77	85.02	77.69	55.50
BLC	69.13	88.13	87.14	77.61	57.14
FLC	69.84	88.24	86.88	79.79	57.01
PL	72.60	90.75	89.06	82.53	57.59
F-div	73.09	91.64	89.70	82.53	57.10
LS (best)	73.44	91.57	89.80	82.76	55.84
NLS (best)	74.24	91.97	90.29	82.99	58.59

- On real world noise benchmarks negative smoothing is strong among simple baselines.
- On Clothing1M and CIFAR noisy sets negative smoothing ranks highly without extra complexity.
- The method is **simple to adopt** and pairs well with advanced robust training.

19

Limitations and side-effects: overconfidence risks

- Overly negative smoothing can overfit clean data and harm calibration.
- Guardrails include warm up bounding the magnitude and close validation monitoring.
- When noise is low revert to label smoothing or vanilla loss to avoid **sharp overconfidence**.

- Use label smoothing at low noise and switch to negative smoothing as noise increases.
- Negative smoothing is simple effective and closely connected to robust loss designs.
- **Future work** extends negative labels beyond classification for [broader impact](#).

- The authors study how smoothing behaves under noisy labels and how to set the rate.
- The problem is to balance confidence on correct patterns against mislabeled data.
- The aim is [better generalization](#) with a simple choice of **smoothing rate**.

Thank You

Questions & Discussion